

EVALUATING TOXIC CHEMICAL EXPOSURE IN E-CIGARETTE USE

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Executive Summary

A recent Australian media report titled “*New Australian vaping research finds ‘suite of chemicals’ in liquids used in vapes, some at ‘dangerously high’ levels*” highlights potential toxicological risks associated with e-cigarette use. The article is based on a study analysing 65 commercially available vape liquids, including products marketed as nicotine-free, to assess their chemical composition and potential health impacts.

The study identified a wide range of chemical substances, including polycyclic aromatic hydrocarbons (PAHs), benzaldehyde, benzyl alcohol, ethyl vanillin, and residues of pesticides and disinfectants such as 2-chlorophenol. Several of these compounds are known to be irritants or have carcinogenic potential. Vapes contain a metal heating coil composed of chromium, nickel and iron that degrades over time with use, leading to the detection of low levels of these heavy metals in the products. In addition, possible plastic degradation products were detected, likely originating from device components. The research also demonstrated that repeated heating of vape liquids can alter chemical composition, potentially generating additional toxic by-products. Notably, nicotine was detected in some products labelled as nicotine-free, raising concerns regarding product regulation and labelling accuracy.

The findings presented in the media article are broadly supported by the evidence, particularly regarding the presence of potentially harmful chemicals and the variability in product composition. However, the claim that certain chemicals are present at “dangerously high” levels requires further evaluation in the context of exposure dose, inhalation toxicology, and frequency of use. Overall, the article raises justified concerns about the potential health risks of vaping and highlights the need for stricter regulatory oversight and further toxicological investigation into inhalation exposure from e-cigarette products.

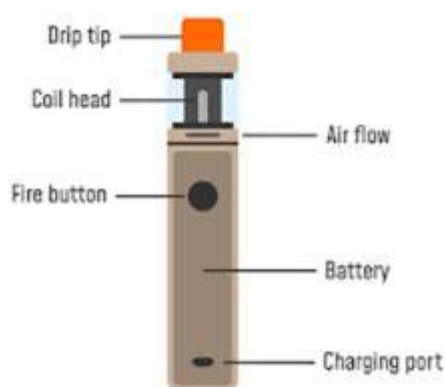


Fig: A vape (CBD Oil King, 2023)

Overview

The selected media article reports, published by ABC news in 2021, reports Australian research identifying a “suite of chemicals” in commercially available 65 vape liquids, including substances described as potentially harmful or present at “dangerously high” levels.

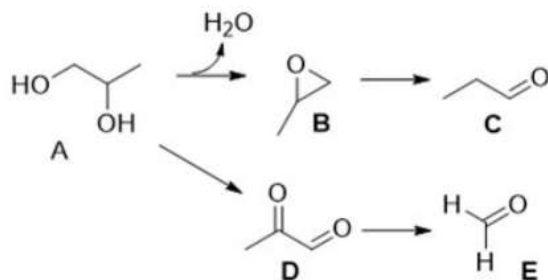
The issue is toxicologically significant because vaping involves direct inhalation of aerosolised chemicals into the respiratory tract (airways and lungs) (Eaton, Kwan and Stratton, 2018). Broader scientific evidence supports the concern that e-cigarette aerosol is not harmless water vapour and may contain nicotine, cancer-causing chemicals, volatile organic compounds, flavouring agents, heavy metals and particles that can be inhaled deep into the lungs. This is important because the lung is a sensitive exposure route, and chemicals that may be acceptable in food are not automatically safe when inhaled (Shaw et al., 2026).

The key toxicological issue selected from the article is the **potential carcinogenic risk associated with inhalation exposure to chemicals detected in vape liquids**.

Long-term cancer risk from vaping is difficult to study because cancer may take years or decades to develop, and vaping is relatively new compared with conventional cigarette smoking (Shaw et al., 2026). Therefore, researchers often estimate potential carcinogenic risk using chemical analysis of vape liquids and aerosols, simulated inhalation exposure studies, animal studies, and in vitro (cell-based experiments outside the body) experiments.

Vape liquids commonly contain four main components: propylene glycol, glycerine/glycerol, nicotine, and flavouring agents. Propylene glycol and glycerine act as solvents and aerosol-forming agents, nicotine is included for its addictive effects (CDC, 2024), and flavouring agents are added to improve taste and consumer appeal. Nicotine in vapes is often present as a benzoate salt (Nicotine salt can vaporise at lower temperatures), which can feel smoother to inhale than regular free-base nicotine, making it easier for people to inhale higher nicotine levels (Shaw et al., 2026).

The vape heating coil can reach about 157–266°C with a usual vaping temperature around 190°C (Shaw et al., 2026). Heating during vaping can cause thermal degradation, which means heat breaks down chemicals into new chemical substances. In e-cigarettes, ingredients such as Propylene glycol and glycerine can form reactive carbonyl compounds and aldehydes, including formaldehyde, acetaldehyde and acrolein, under heating and aerosolization conditions (Shaw et al., 2026). Formaldehyde is classified by IARC (International Agency for Research on Cancer) as a human carcinogen while acetaldehyde is classified as possibly carcinogenic to humans. These aldehydes are important because they can irritate respiratory tissue, damage lung cells, react with proteins, and contribute to DNA damage or inflammatory responses. If this DNA damage is not repaired correctly, mutations may occur, increasing the risk of cancer over long-term exposure.

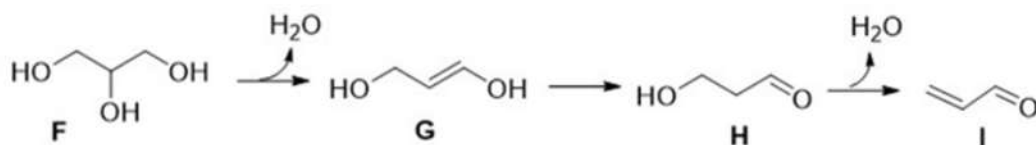


(Shaw et al., 2026)

Fig 1: Thermal degradation of Propylene glycol

Propylene glycol (A) → propylene oxide (B) → propionaldehyde (C)

Propylene glycol (A) → methylglyoxal (D) → formaldehyde (E)



(Shaw et al., 2026)

Fig 2: Thermal degradation of Glycerine

Glycerine (F) → 1,3-dihydroxypropene (G) → 3-hydroxypropanal (H) → acrolein (I)

Flavouring compounds are a key toxicological concern in e-cigarette liquids because they are added for taste but may also influence the toxicity of the inhaled aerosol. Many flavouring additives are themselves aldehydes, such as benzaldehyde, cinnamaldehyde, vanillin and ethyl vanillin, and some contain unsaturated bonds that may make them more chemically reactive when heated. Earlier studies mostly examined the direct evaporation of flavouring compounds such as benzaldehyde and diacetyl, but less attention has been given to whether these flavourings break down during heating. The contribution of flavouring chemicals to toxic aldehyde formation has been less studied because they are usually present in smaller amounts than the main vape liquid solvents, such as propylene glycol and glycerol (Khlystov and Samburova, 2016).

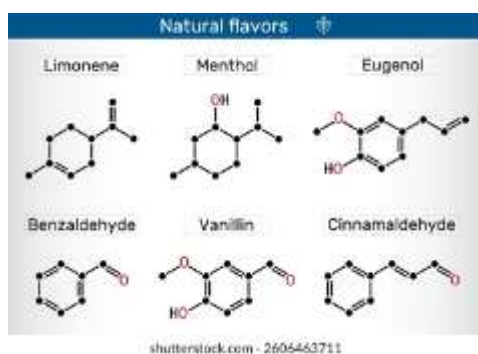


Fig: Flavouring chemicals

PAHs are produced in trace amounts during thermal decomposition of organic material, they are toxicologically important by-products because many act as procarcinogens that undergo metabolic activation by cytochrome P450 enzymes, forming reactive metabolites that can bind to DNA and create DNA adducts. If these DNA adducts are not repaired properly, DNA replication may introduce permanent mutations, which can contribute to carcinogenesis after repeated or chronic exposure (Xue and Warshawsky, 2005).

Polycyclic Aromatic Hydrocarbons

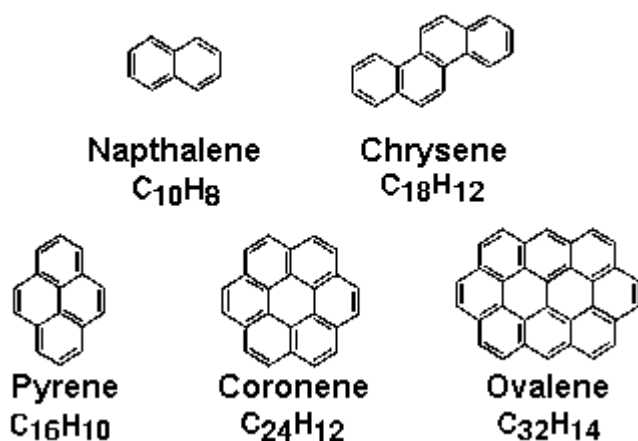
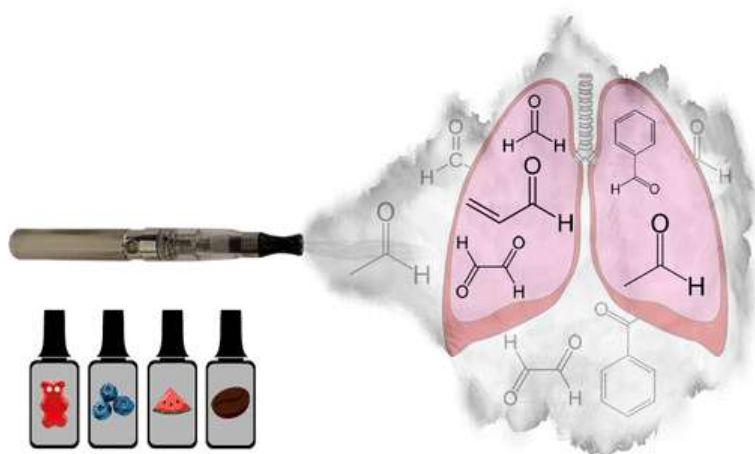


Fig: structure of PAHs

Vaping is often repeated over time. Cancer risk usually relates to chronic exposure, not just a single exposure event. Therefore, the relevant parameters to investigate include the route of exposure, concentration of carcinogenic or potentially carcinogenic chemicals, frequency of vaping, duration of use, ability of chemicals to reach lung tissue, and whether the exposed person belongs to a vulnerable group. Potentially affected groups include regular vape users, adolescents and young adults, former smokers using e-cigarettes, and bystanders exposed to second-hand aerosol; the CDC (Centers for Disease Control and Prevention) notes that bystanders can inhale aerosol exhaled by vape users.



(Khlystov and Samburova, 2016)

Methodology

The investigation was conducted as a literature-based (using peer-reviewed journal articles, toxicology reviews, public health reports, regulatory and health agency documents, cancer classification sources such as IARC) toxicological assessment using the selected ABC News media article as the initial case study. The article was reviewed to identify the main toxicants of concern and the toxicological issue requiring further analysis. Although the article raised several concerns, the methodology focused primarily on evaluating the potential carcinogenic risk associated with inhalation exposure to chemicals detected in vape liquids and vape aerosols.

Toxicological data:

Toxicological Parameter	Data
Exposure concentration	• Typical mixtures often contain about 30–50% glycerol, with the balance largely propylene glycol. Average amount of toxicants delivered per vape puff: Formaldehyde: 7.8 µg/puff (Shaw et al., 2026) Acetaldehyde: 0.02–22.5 µg/puff and 7.71–18.0 µg/puff (2 ranges available) (Shaw et al., 2026) Acrolein: 0.012–1.37 µg/puff and 12.4–19.2 µg/puff (2 ranges available) (Shaw et al., 2026). • Vape liquid mixtures often contain about ~1–4% flavour chemicals (Tierney et al., 2024). • Total flavour chemical concentrations range from 0.005–362 mg/mL; about 85% are >1 mg/mL and 37% are >10 mg/mL (Tierney et al., 2024). (Strong evidence that flavour chemical concentration varies widely and can be high). • Nicotine concentrations vary widely between products, with studies analysing liquids labelled around 6–22 mg/mL and experimental studies often using concentrations such as 18 mg/mL (Peace et al., 2016).

Carcinogenic (IARC) classification	Formaldehyde: Group 1: carcinogenic to humans. Acrolein: Group 2A - Probably carcinogenic to humans - There's strong evidence that the substance can cause cancer, but the evidence in humans may not be fully conclusive. There is usually enough evidence from animal studies and/or strong biological evidence. Acetaldehyde: Group 2B - Possibly carcinogenic to humans - There's some evidence that the substance may cause cancer, but the evidence is more limited or less certain than Group 2A. Flavouring chemicals: Not classified - IARC states that for agents not classified, no conclusion about non-carcinogenicity or overall safety should be assumed.
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Mechanism of toxicity:

Formaldehyde

It is highly reactive and can form DNA-protein crosslinks, DNA-DNA crosslinks and DNA adducts. These damaged sites can block DNA replication and interfere with cell repair processes (DL, LY and K, 2018).

In inhalation exposure, a major mechanism is local cytotoxicity in the respiratory epithelium, especially nasal/airway tissue. When cells are repeatedly injured or killed, the tissue responds by increasing regenerative cell growth. This increases the chance that DNA errors become fixed as mutations (Blouin and Saini, 2024).

Acetaldehyde

It forms DNA adducts that are linked with mutations, DNA strand breaks, DNA-DNA crosslinks and sister chromatid exchanges. Acetaldehyde can also form protein adducts, which may impair enzymes involved in DNA repair and cellular defence. It can contribute to oxidative stress (happens when the body produces too many free oxygen radicals and does not have enough antioxidants to neutralise the) and lipid peroxidation (free radicals attack the fatty parts of cell membranes) products, creating additional reactive aldehydes that further damage cells (Mizumoto et al., 2017).

Acetaldehyde can also form DNA-DNA crosslinks and contribute to double-strand breaks. These are serious because double-strand breaks affect both strands of DNA and can lead to chromosomal instability if repaired incorrectly (Mizumoto et al., 2017).

Acrolein

Acrolein can damage DNA, interfere with DNA repair, forms DNA adducts, increase oxidative stress, and cause chronic inflammation, alters normal cell growth and cell death. Acrolein can also deplete glutathione, an important antioxidant that helps protect cells, making cells more vulnerable to further chemical damage (Liu, Zhu and Xie, 2010).

Flavouring chemicals

Flavouring chemicals may increase the amount of toxic aldehydes formed during vaping, by thermal breakdown (Shehata et al., 2023).

Some flavouring chemicals suppress cilia movement in the airways, reducing the lungs' ability to clear toxic particles and increasing the risk of respiratory infections, leading to lung cancer (Shehata et al., 2023).

PAHs

PAHs are often procarcinogens, meaning the parent compound is not always the most DNA-reactive form. After inhalation, PAHs can enter airway/lung cells because they are lipophilic. Inside the body, enzymes such as cytochrome P450, can metabolically activate PAHs into more reactive electrophilic metabolites (Xue and Warshawsky, 2005). One major pathway is:

PAH $\rightarrow$ epoxide $\rightarrow$ dihydrodiol $\rightarrow$ diol epoxide (Xue and Warshawsky, 2005)

The diol epoxide form is highly reactive and can bind covalently to DNA bases, especially guanine and sometimes adenine, forming DNA adducts (Xue and Warshawsky, 2005).

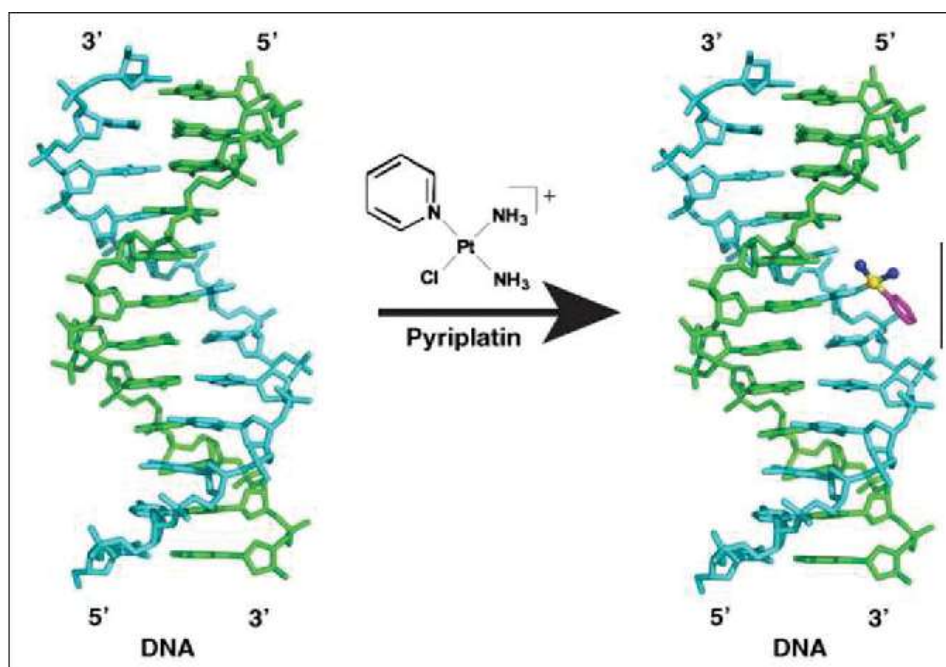


Fig: DNA adduct

(Semanticscholar.org, 2026)

Justification

- Mechanisms of toxicity were included to explain how chemicals detected in vape liquids may cause biological harm, particularly through pathways linked to carcinogenesis.
- Average toxicant delivery per puff was included to estimate the realistic inhalation exposure dose a vape user may receive.
- IARC classifications were included to identify whether the chemicals detected in vape liquids or aerosols are recognised as known, probable, or possible carcinogens, which helps assess the credibility of the article's cancer-risk concerns.
- Frequency and duration of exposure are also considered because cancer risk is generally associated with repeated or chronic exposure rather than a single exposure event. In vaping, exposure depends on how often a person vapes, how many puffs are taken, how long the behaviour continues, and the concentration of chemicals produced by the device and liquid.
- LD₅₀ and LC₅₀ values were not prioritised because they mainly measure acute lethality, whereas this report focuses on long-term carcinogenic risk from repeated inhalation exposure.

Discussion

Dangerously high levels?

Article phrases:

“.....dangerously high levels of lung irritant benzaldehyde, which is added to vapes to give them an almond flavour, in 61 of the 65 samples”

“.....high levels of a cinnamon flavouring known as trans-cinnamaldehyde in up to 48 of the liquids”

- The “dangerously high” claim is partially justified because benzaldehyde and trans-cinnamaldehyde are flavouring chemicals, and published studies show that flavouring chemical concentrations in vape liquids vary widely, but lower concentrations appear to be more common than very high concentrations.
- A chemical concentration may appear high in the liquid, but the actual risk depends on how much is transferred into the aerosol and inhaled per puff.

PAH exposure?

Article phrase:

“There's not really any safe level of exposure to PAH that's accepted”

- Claim is justified, particularly because several PAHs are known as procarcinogens. Although PAHs may be present only at trace levels in vape liquids, their detection remains important because carcinogenic compounds can be relevant even at low concentrations, especially under repeated or chronic inhalation exposure.

Relevance of mechanism of toxicity to tumour/cancer?

Normally, the body has repair systems that detect and fix DNA damage. However, if exposure happens repeatedly, the amount of damage may become too great, or repair may be incomplete or incorrect. When damaged DNA is copied during cell division, mistakes can become permanent mutations. These mutations are important because cancer often begins when mutations affect genes that control cell growth, cell division, DNA repair or programmed cell death.

Over time, some cells may collect mutations that give them abnormal advantages. For example, they may start dividing faster, ignore signals to stop dividing, avoid programmed cell death, or repair DNA less effectively. These abnormal cells can survive longer than they should and multiply. As more mutations accumulate, the cells become increasingly uncontrolled, eventually forming a tumour.

Toxicant exposure from vaping is highly variable. The amount of chemicals generated or delivered in vape aerosol can depend on:

Coil voltage — higher voltage generally increases coil temperature, which may increase thermal breakdown of vape liquid components and produce more toxic by-products.

Device type — different devices have different power outputs, coil designs, airflow systems and heating efficiency, which can change the chemical composition of the aerosol.

Liquid composition — the ratio of propylene glycol to glycerine, nicotine content, additives and contaminants can affect which toxicants are formed and in what amounts.

Flavouring chemicals — different flavours contain different chemical structures, and some flavouring compounds may evaporate into the aerosol or break down into reactive aldehydes during heating.

Puffing behaviour — longer, deeper or more frequent puffs can increase the amount of aerosol inhaled and may also increase coil temperature, raising exposure to toxicants.

Device age and coil condition — older or degraded coils may release more metals or heat unevenly, potentially altering toxicant formation.

Frequency and duration of use — repeated vaping over months or years increases cumulative exposure, which is especially important for long-term endpoints such as carcinogenicity.

- These information makes it difficult to define a single “safe” or “dangerous” exposure level and supports the need for product-specific toxicological assessment.

Conclusion

The media article is justified in raising toxicological concern about chemicals detected in vape liquids, particularly in relation to potential long-term carcinogenic risk. The presence of PAHs, reactive aldehydes such as formaldehyde, acetaldehyde and acrolein, and flavouring chemicals with uncertain inhalation safety supports the need for careful risk assessment. These substances are relevant because they may contribute to cancer-related processes such as DNA damage, oxidative stress, chronic inflammation, impaired repair and repeated respiratory tissue injury.

However, the severity of risk cannot be stated absolutely from chemical presence alone. The article's concern is scientifically reasonable, but terms such as "dangerously high" should be interpreted cautiously and supported by exposure data and inhalation-specific toxicity values. Overall, the findings support stronger regulation, clearer product labelling and further long-term research into the health effects of vaping.

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